Yash Verma

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Professional Summary

AI/ML engineer and former Ford Senior Analyst with nearly four years of experience developing Python automation, NLP-based security analytics, GCP/BigQuery data pipelines, and enterprise decision-support solutions. M.S. Computer Science (AI/ML) candidate with hands-on experience across data preprocessing, feature engineering, model training and evaluation, MLOps, API deployment, Generative AI, RAG, and agentic workflows.

Technical Skills

- **Programming / Data:** Python, SQL, Java, JavaScript, Pandas, NumPy, PySpark
- **Machine Learning:** PyTorch, TensorFlow, scikit-learn, OpenCV, classification, NLP/NER, computer vision, feature engineering, model evaluation and tuning
- **Generative AI:** LLMs, RAG, Agentic AI, LangChain, ChromaDB, Ollama, embeddings, semantic retrieval, retrieval evaluation
- **Cloud / Analytics:** Docker, CI/CD Pipelines, GCP BigQuery, AWS, QlikView, REST APIs, STIX/TAXII
- **MLOps / Engineering:** MLflow, DVC, GitHub Actions, Docker, FastAPI, Git, CI/CD

Experience

Research Assistant — **May 2026 – Present**
UB Safe and Efficient Autonomous Systems (SEAS) Lab — *Buffalo, NY*

- Trained and tuned LECSFormer for crack detection on dashcam images using semi-automated annotation, preprocessing, custom augmentation, and adaptive cropping.
- Improved F1 score from 0.29 to 0.55 and precision from 0.20 to 0.70 through model and data-pipeline iteration.
- Integrated inference into a ROS2/C-V2X vehicle pipeline combining OpenCV capture, camera calibration, RTK GPS, and GPS-to-pixel projection for targeted crack detection.
- Implemented OBU/RSU WSMP communication using Commsignia pycmssdk, including metadata transfer, image chunking, feedback reconstruction, and timing instrumentation.

Senior Analyst (Cybersecurity Analytics & Automation) — **Sep 2021 – Aug 2025**
Ford Motor Company — *Chennai, India*

- Built production Python and Docker-based analytical, automation, and data-ingestion workflows for enterprise threat intelligence operations, reducing manual processing overhead by 40%.
- Trained an NLP named-entity recognition model on cybersecurity PDF reports to extract IP addresses, domains, URLs, and file hashes; improved entity-detection performance by 25% over the previous workflow and evaluated precision, recall, F1, and false-positive rate.
- Built STIX/TAXII and REST API integrations across MISP, ThreatConnect, Mandiant, and Auto-ISAC data sources, reducing rework from 20% to 5% and improving queue age by 40–60%.
- Developed PySpark and SQL pipelines to transform GCP BigQuery data powering 10 QlikView cybersecurity dashboard views, enabling teams, directors, and senior leaders to monitor operational and risk KPIs and prioritize remediation.
- Collaborated with product owners, engineers, analysts, and stakeholders in Agile teams; documented technical designs, ML experiments, data workflows, and operational results.

Selected AI/ML Projects

Agentic Technical Documentation Assistant [GitHub] — **Python, Ollama, LangChain, ChromaDB**

- Built a privacy-preserving agentic RAG assistant for technical API/SDK documentation, indexing 746 document chunks and returning grounded answers with source citations using local LLMs.
- Evaluated top-k retrieval and instrumented retrieval scores, latency, token usage, tool-call outcomes, refusal guardrails, and failure cases to improve grounding and diagnose errors.

Distributed Deep Learning Benchmark — **PyTorch, DDP, CUDA, NVIDIA L40S**

- Built CPU, single-GPU, and two-GPU ResNet-34 training pipelines with PyTorch DDP, `torchrun`, DistributedSampler, and mixed precision.
- Achieved a 1.7× speedup on CIFAR-10, reducing training time from 45.53 seconds on one GPU to 26.8 seconds on two GPUs while tracking throughput and evaluation metrics.

Customer Churn MLOps Pipeline [GitHub] — **scikit-learn, MLflow, DVC, GitHub Actions**

- Built a churn prediction pipeline with feature engineering and model comparison, achieving 0.832 ROC-AUC, 0.605 F1, and 0.80 recall on held-out test data.
- Used DVC, MLflow, and GitHub Actions for reproducibility, model tracking, testing, and quality gates; served predictions with FastAPI and Docker.

Education

University at Buffalo (SUNY Buffalo), Buffalo, NY — **Dec 2026**
Master of Science in Computer Science (AI/ML) — GPA: 3.7 / 4.00

Sant Longowal Institute of Engineering and Technology (SLIET), Punjab, India — **June 2021**
Bachelor of Engineering in Computer Science — GPA: 9.18 / 10